

All-Degree Rigidity of a Weighted Keller Class

Classification, degenerations,
and an obstructed double point

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Abstract

We classify, without any bound on ordinary degree, a class of polynomial Keller maps in three variables naturally singled out by the Jacobian project in the `ProveIt` repository. The source weights are $(-1, 1, 2)$ and the target weights are $(2, 1, -1)$. In invariant coordinates $t = xy$ and $v = x^2z$, we assume that the first two numerators are affine in v and that the third numerator is affine in (t, v) . Every normalized map is either an explicitly invertible tame map or a member of a two-parameter diagonal orbit of the known Alpöge map. In the latter case its degree profile is forced to be $(7, 6, 4)$ and its expanded support has exactly sixteen monomials.

The proof replaces a degree-bounded coefficient search by one-variable polynomial rigidity. An equivariant source shear extends the classification to arbitrary polynomial dependence of r on t when its v -coefficient is constant, and gives the exact noninvertible degree spectrum $\{7, 8, 10, 12, \dots\}$. We also identify the reduced closure of the nonconstant branch as an affine plane whose coordinate axes are precisely its invertible members. Finally, for the normalized degree-seven coefficient problem with third numerator $1 + t + v$, we prove that the entire coefficient scheme is a double point, $\text{Spec } \mathbb{Q}[\varepsilon]/(\varepsilon^2)$. An explicit nonzero infinitesimal deformation is obstructed at second order. All finite identities and obstruction certificates are independently reproduced by an included exact-arithmetic script.

Status. Human-readable proofs and reproducible symbolic certificates; not a Lean/Rocq formalization or a peer-reviewed publication. The results extend the inspected repository statements, but publication priority is not established. This report does not claim a new counterexample to the Jacobian conjecture or a classification of unrestricted Keller maps.

[write] Batch 36, manuscript 01, placed in `1a1396d4d`: a report of ProveIt’s research-report collection that continues the formal project `Algebra/JacobianConjecture`. AI-assisted, unrefereed, not formalized; labels carry the prefix `wkr`:. The weighted determinant identity and the collision mechanism are the project’s own (Section 1.4); Gao’s three-dimensional map has *geometric* degree four (Section 1.5). Provenance: Appendix D.

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1 Research question, provenance, and contribution

A useful research question should distinguish a structural theorem from a large but finite search. The Jacobian component of `ProveIt` provides a particularly concrete opportunity: it gives explicit maps, exact collision certificates, a weighted determinant reduction, and degree-bounded searches for simpler representatives. We ask whether one of its natural coefficient classes is rigid *without imposing a degree bound*. We then ask what remains of the apparent uniqueness when coefficients are allowed to contain nilpotents. These are separate questions: a unique field-valued solution need not be a reduced point of its coefficient scheme.

1.1 The repository baseline

The inspected snapshot is commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

The relevant files are the Jacobian project README, its research README, and `verify_equivariant_sparsity.py`; immutable references are supplied in the bibliography [1, 2, 3].

The repository's three-variable map can be written

$$\begin{aligned} u &= 1 + xy, & h &= u^2 z + y^2(1 + 3u), \\ F_0(x, y, z) &= (uh, y + 3xh, x(5 - 3u - x^2 z)). \end{aligned} \tag{1.1}$$

Its Jacobian determinant is -2 , and

$$F_0(-1, 1, 5) = F_0(0, -2, -16) = (0, -2, 0). \tag{1.2}$$

These identities, their formal verification, and their provenance are already part of the repository [1]. We verify the displayed identities again in the companion script but do not present them as new.

The research files restrict a sparsity calculation to maps of ordinary degree at most seven whose invariant numerators p, q are affine in v and whose third numerator r is affine in (t, v) . After normalizing $r = 1 + t + v$, the script proves that eleven optional coefficients cannot vanish, and identifies the branch $q_{40} = 0$ with the normalized map. Its certificate contains a square $(q_{21} - 4)^2$, but the repository discussion does not identify the entire coefficient scheme as the dual numbers [2, 3].

1.2 What this article establishes

The main result, Theorem 3.1, removes the ordinary degree restriction from that structural class. It also handles constant r and proves that the two one-term nonconstant cases are impossible. The proof forces the normalized nonconstant map outright, so it proves $q_{40} = 0$ rather than imposing it.

Theorem 6.3 extends the result beyond affine r by an explicit triangular normal form. Its degree-spectrum corollary determines all possible maximum degrees in that larger class. Theorem 7.1 describes the reduced two-parameter closure and its invertible boundary. Theorem 8.1 strengthens uniqueness in the fixed normalized degree-seven slice to an exact scheme presentation, and Theorem 8.4 identifies the full scheme on the nonconstant-parameter open locus. These results connect three levels of information:

Level	Conclusion proved here
Field-valued maps	An all-degree automorphism/counterexample dichotomy.
Reduced families	No deformation after fixing $r = 1 + t + v$; a parameter plane before fixing its two nonconstant coefficients.
Nonreduced families	One square-zero parameter in the degree-seven slice, with a nonzero second-order obstruction.

1.3 Relation to contemporary literature and limits

Shaska studies graded Keller maps, their quotient geometry, and the role of the sign pattern of weights [4]. Gao gives a tangent-sweep construction and further explicit examples, including a three-dimensional map of degree four [5]. Our subject is narrower: rigidity within a specified invariant-numerator class, not the construction of the known map or a minimum-degree theorem for all three-variable maps. In particular, our forced degree seven is compatible with examples outside this class of lower degree.

The searched sources and the inspected repository justify the comparison above, not a universal priority claim. The mathematical contribution of this report is its stated and proved classification and deformation results. Whether equivalent formulations already occur elsewhere requires further bibliographic review. Nothing in the proofs depends on accepting a new external claim about the general Jacobian problem: the particular map, its determinant, and its collisions are explicit.

1.4 What the project already contains

[write] This subsection and the next were added when the manuscript was written into ProveIt. Two ingredients that the manuscript presents in its own words are already recorded in the project’s research notes [2], `Algebra/JacobianConjecture/Research/README.md`, lines 188–224, at the pin and unchanged at the placement commit:

- the lift $F = (p(t, v)/x^2, q(t, v)/x, x r(t, v))$ with $t = xy$, $v = x^2z$, the polynomiality conditions of (2.3) in their general form, *the same determinant identity* (2.4) with the same chain-rule proof in the localization at x (Proposition 2.2), and the normalization (2.6) with determinant -1 (lines 188–208);
- the collision mechanism: with $\alpha = [t^2]p$ (the notes write a , which is a family parameter here), $F(0, y, z) = (\alpha y^2 + z, y, 0)$ and $F(1, t_0, v_0) = F(0, q(t_0, v_0), p(t_0, v_0) - \alpha q(t_0, v_0)^2)$ whenever $r(t_0, v_0) = 0$ (lines 210–224). The collision (6.6) is this mechanism at $t_0 = 0$, $v_0 = -1/\gamma$.

The manuscript cites the research notes for its baseline but not at these two points; they are credited here. What is new relative to the project is the all-degree classification and everything after it: the project’s sparsity certificate (`verify_equivariant_sparsity.py`, described in lines 240–248 of the notes) covers ordinary degree at most seven with $q_{40} = 0$ left as the “remaining branch”, and says it is “deliberately not advertised as a global sixteen-monomial theorem”. Corollary 6.1 proves $q_{40} = 0$ and the support profile in all degrees, inside the class only. For constant r the notes invoke Moh’s plane theorem through degree 100 (lines 232–238), in a class that allows v^2 terms; case (A) proves tameness in all degrees without Moh, but only for p, q affine in v , so the two statements sit side by side.

Formal status. The project’s Lean and Rocq/Coq developments formalize the map F_0 (Lean `counterexample`, `Algebra/JacobianConjecture/Lean/JacobianConjecture/C`

ounterexample.lean, lines 95–101), its determinant -2 (jacobianDet_counterexample), the collision (1.2) (collision), the weight- $(-1, 1, 2)$ torus action (counterexample_scaling, Algebra/JacobianConjecture/Lean/JacobianConjecture/Scaling.lean), the mirror symmetry and the refutation of the three-dimensional conjecture; the Coq development proves the corresponding statements. None of the weighted determinant identity, the normalization (3.4), the family $F_{a,b}$ or any theorem of this report is formalized.

1.5 Gao’s three-dimensional map has geometric degree four

[write] The “three-dimensional map of degree four” of [5] cited above and in Section 10 is Gao’s map G of his Theorem 3.5, and its *degree four is the geometric degree*, the number of preimages of a generic point, the sense of “degree” defined in his abstract. Its three components have ordinary degrees 4, 11 and 12 and its Jacobian determinant is 2: this was read from his Section 3.5 at the write and checked there with SymPy 1.14.0 from his printed formulas (component degrees 4, 11, 12 and $\det J_G = 2$ as polynomial identities). So G is *not* a three-dimensional Keller map of lower ordinary degree than seven. The sentence “our forced degree seven is compatible with examples outside this class of lower degree” remains true as a compatibility statement, but G is not such an example; for Alpöge’s map Gao’s Theorem 3.3 gives geometric degree three. The project’s framing is unaffected: its notes say that the finite searches “do not prove that ordinary degree seven or sixteen monomials is globally minimal in dimension three” and that a simpler three-variable map would have degree at least four and, below seven, would break the weighted symmetry (Algebra/JacobianConjecture/Research/README.md, lines 12–15 and 262–269). Whether such a map exists is not settled by any source cited here.

2 Weighted coordinates and the exact determinant reduction

Throughout the classification, K is an arbitrary field of characteristic zero. All polynomial equalities are formal equalities over K .

Definition 2.1. A polynomial map $F : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is a *Keller map* if its formal Jacobian determinant is a nonzero element of K . We use the source action

$$(x, y, z) \mapsto (\rho^{-1}x, \rho y, \rho^2 z)$$

and the target action with weights $(2, 1, -1)$. Equivariance means that the three coordinate polynomials are weighted homogeneous of weights 2, 1, -1 , respectively.

2.1 Invariant numerators

Set

$$t = xy, \quad v = x^2 z. \tag{2.1}$$

Both are invariant under the source action. Every equivariant polynomial map has a unique expression in the localization at x of the form

$$F = \mathcal{L}(p, q, r) := \left(\frac{p(t, v)}{x^2}, \frac{q(t, v)}{x}, xr(t, v) \right), \quad p, q, r \in K[t, v]. \tag{2.2}$$

For example, a monomial $x^i y^j z^k$ of weight 2 satisfies $i = j + 2k - 2$; multiplying it by x^2 gives $t^j v^k$. The corresponding formulas for weights 1 and -1 are $i = j + 2k - 1$ and $i = j + 2k + 1$. They prove existence and uniqueness of the numerators monomial by monomial.

For general numerators, polynomiality requires that every monomial $t^i v^j$ in p satisfy $i + 2j \geq 2$, and every such monomial in q satisfy $i + 2j \geq 1$. In the class studied here, $\deg_v p, \deg_v q \leq 1$, so these conditions reduce exactly to

$$p(0, 0) = p_t(0, 0) = q(0, 0) = 0. \quad (2.3)$$

There is no additional restriction on r for polynomiality. The apparent fractions in (2.2) are therefore polynomial expressions whenever (2.3) holds in our class.

Proposition 2.2 (Weighted determinant identity). *For a polynomial lift (2.2),*

$$\det J_F = \mathcal{J}(p, q, r) := \det \begin{pmatrix} -2p & p_t & p_v \\ -q & q_t & q_v \\ r & r_t & r_v \end{pmatrix}, \quad (2.4)$$

where the right side is evaluated at $(t, v) = (xy, x^2z)$.

Proof. Compute in coordinates (x, t, v) on $x \neq 0$. The coordinate change $(x, y, z) \mapsto (x, xy, x^2z)$ has determinant x^3 . The derivative matrix of $(x^{-2}p, x^{-1}q, xr)$ with respect to (x, t, v) has rows

$$(-2x^{-3}p, x^{-2}p_t, x^{-2}p_v), \quad (-x^{-2}q, x^{-1}q_t, x^{-1}q_v), \quad (r, xr_t, xr_v).$$

Factoring x^{-2}, x^{-1}, x from its rows and x^{-1} from its first column leaves the matrix in (2.4) and contributes x^{-3} . The chain rule cancels this with x^3 . Localization is injective on $K[x, y, z]$, so the identity holds as a polynomial identity, including on $x = 0$. $\square$

[write] This identity, with this proof, and the normalization below are already in the project's research notes (Section 1.4).

2.2 Normalization is exhaustive

Suppose that (2.3) holds and $\mathcal{J}$ is a nonzero constant. Evaluating at $(t, v) = (0, 0)$ gives

$$\mathcal{J}(0, 0) = -p_v(0, 0)q_t(0, 0)r(0, 0). \quad (2.5)$$

All three factors are therefore nonzero. Dividing the three target coordinates by these factors normalizes

$$p_v(0, 0) = q_t(0, 0) = r(0, 0) = 1. \quad (2.6)$$

For a Keller map, the resulting constant determinant is -1 . This operation preserves the structural restrictions on p, q, r , ordinary degrees, support sizes, and invertibility or noninvertibility. Thus a classification in this normalization classifies the unnormalized maps up to independent nonzero target scalings. We can write

$$r = 1 + \beta t + \gamma v. \quad (2.7)$$

The word *normalized* below always includes (2.3) and (2.6); it does not implicitly assume $\beta = \gamma = 1$.

3 The complete all-degree classification

Define the following polynomials over $\mathbb{Q}[a, b, t, v]$:

$$p_{a,b} = v - \frac{8a}{9}t^2 - 2abtv + \frac{28a^2b}{27}t^3 + \frac{4a^2b^2}{3}t^2v - \frac{8a^3b^2}{27}t^4 - \frac{8a^3b^3}{27}t^3v, \quad (3.1a)$$

$$q_{a,b} = t + 9bv - 8abt^2 - 12ab^2tv + 4a^2b^2t^3 + 4a^2b^3t^2v, \quad (3.1b)$$

$$r_{a,b} = 1 + abt + ab^2v. \quad (3.1c)$$

Let $F_{a,b} = \mathcal{L}(p_{a,b}, q_{a,b}, r_{a,b})$.

Theorem 3.1 (All-degree classification). *Let $p, q, r \in K[t, v]$ satisfy $\deg_v p, \deg_v q \leq 1$, let r be affine in (t, v) , and suppose that their lift is polynomial and normalized as in (2.6). Then $\mathcal{J}(p, q, r) = -1$ if and only if exactly one of the following descriptions applies.*

(A) $r = 1$ and, for unique $\lambda \in K$ and $B \in t^2K[t]$,

$$p = v + B(t), \quad q = t + \lambda(v + B(t)). \quad (3.2)$$

Its lift is a tame polynomial automorphism.

(B) r is nonconstant, both β and γ in (2.7) are nonzero, and

$$(p, q, r) = (p_{a,b}, q_{a,b}, r_{a,b}), \quad a = \frac{\beta^2}{\gamma}, \quad b = \frac{\gamma}{\beta}. \quad (3.3)$$

Its lift is noninjective. Its coordinate degrees are $(7, 6, 4)$, and its coordinate support sizes are $(7, 6, 3)$.

In particular, there are no normalized Keller maps in this class with exactly one of β, γ nonzero. No ordinary degree bound is assumed.

We prove the classification in Sections 4–5. The invertibility, collisions, and support assertions are proved in Section 6. The forward direction is the substantive rigidity statement; the reverse direction follows either from the formulas there or by substitution into (2.4).

3.1 The normalized representative

Put $F_* = F_{1,1}$. Its numerators are

$$p_* = v - \frac{8}{9}t^2 - 2tv + \frac{28}{27}t^3 + \frac{4}{3}t^2v - \frac{8}{27}t^4 - \frac{8}{27}t^3v, \\ q_* = t + 9v - 8t^2 - 12tv + 4t^3 + 4t^2v, \quad r_* = 1 + t + v.$$

It is exactly a normalization of (1.1):

$$F_* = \text{diag}\left(-\frac{1}{2}, -\frac{3}{2}, \frac{1}{2}\right) \circ F_0 \circ \text{diag}\left(1, -\frac{2}{3}, -2\right). \quad (3.4)$$

Thus the nonconstant part of the classification identifies a known representative rather than producing an unrelated counterexample.

4 One-variable rigidity lemmas

The decisive step is that constant weighted Wronskians control degree even when arbitrarily high-degree polynomials are allowed initially.

Lemma 4.1 (A degree bound from a constant weighted Wronskian). *Let $B, D \in K[t]$ satisfy*

$$DB' - 2BD' = \kappa, \quad \kappa \in K^\times. \quad (4.1)$$

Then $D \neq 0$, $\deg D \leq 1$, and $\deg B \leq 2$. More precisely, either D is a nonzero constant and B has degree one, or D has degree one and $B = \eta D^2 + \mu$ for some $\eta, \mu \in K$, with $-2\mu D' = \kappa$.

Proof. Neither B nor D can be zero. If D is constant, (4.1) gives $B' = \kappa/D$, so B has degree one.

Now put $n = \deg D \geq 1$ and $m = \deg B$. Unless $m = 2n$, the leading term on the left has degree $m + n - 1$ and nonzero coefficient proportional to $m - 2n$. This uses characteristic zero. If $m > 2n$, it contradicts the constant right-hand side. If $m = 2n$, subtract from B the unique scalar multiple of D^2 that cancels its leading coefficient. This changes neither side of (4.1). The resulting polynomial B_0 is nonzero and has degree $m_0 < 2n$. If initially $m < 2n$, just take $B_0 = B$.

The leading term calculation for B_0 now gives $m_0 + n - 1 = 0$. As $m_0 \geq 0$ and $n \geq 1$, we obtain $n = 1$ and $m_0 = 0$. Hence $B = \eta D^2 + \mu$, and substitution gives $-2\mu D' = \kappa$. $\square$

Lemma 4.2 (A cubic–square relation). *Suppose $A, C \in K[t]$ are nonzero and*

$$2CA' - 3AC' = 0, \quad A(0) = 1. \quad (4.2)$$

Then there are unique $E \in K[t]$ with $E(0) = 1$ and $c \in K^\times$ such that

$$A = E^3, \quad C = cE^2. \quad (4.3)$$

Proof. The derivative of A^2/C^3 is zero. A rational function over a characteristic-zero field with derivative zero lies in K : writing it in coprime numerator–denominator form, the identity $f'g - fg' = 0$ forces each nonconstant numerator or denominator to divide its own derivative, which is impossible by degree. Thus $A^2/C^3 \in K^\times$.

For every irreducible polynomial π , unique factorization gives $2 \operatorname{ord}_\pi A = 3 \operatorname{ord}_\pi C$. Consequently the multiplicities in A are multiples of three and those in C are the corresponding multiples of two. Because $A(0) = 1$, none of the common factors vanishes at zero. Normalize their product to have value one there. The resulting polynomial E satisfies $A = E^3$ and $C = C(0)E^2$. No extraction of a scalar root or extension of K is necessary. Uniqueness follows from the factorization and the value at zero. $\square$

Lemma 4.3 (The remaining common factor is linear). *Suppose that $A(0) = 1$, $B(0) = -1$, and*

$$DB' - 2BD' = -1, \quad (4.4)$$

$$DA' + 2CB' - 3AD' - 2BC' = 0, \quad (4.5)$$

$$2CA' - 3AC' = 0. \quad (4.6)$$

If $C \neq 0$, then in (4.3) the polynomial E has degree at most one.

Proof. Lemma 4.1 bounds $\deg D \leq 1$ and $\deg B \leq 2$. Lemma 4.2 gives $A = E^3$, $C = cE^2$. Suppose $m = \deg E \geq 2$.

If D has degree one, the leading coefficient of $DA' - 3AD'$ in degree $3m$ is $3(m-1)$ times the leading coefficient of D times the cube of that of E . It is nonzero. The other part $2CB' - 2BC'$ of (4.5) has degree at most $2m + 1 < 3m$, a contradiction.

If D is constant, it is nonzero and B has degree one. The term DA' has degree $3m - 1$, whereas $2CB' - 2BC'$ has degree at most $2m$. Again the leading term cannot cancel. Thus $\deg E \leq 1$. $\square$

Remark 4.4. The lemmas address the possible high-degree cancellations explicitly. A coefficient search stopped at any fixed degree would not establish these bounds. Characteristic zero is essential to the leading coefficient arguments and to the assertion about rational functions with zero derivative.

5 Proof of the classification, including exceptional cases

5.1 The case $\gamma \neq 0$: a moving coordinate

First reduce $r = 1 + \beta t + \gamma v$ to coefficient one on v . Replace

$$\tilde{p}(t, v) = \gamma p(t, v/\gamma), \quad \tilde{q}(t, v) = q(t, v/\gamma), \quad \tilde{r}(t, v) = r(t, v/\gamma). \quad (5.1)$$

The determinant and normalization are unchanged, while $\tilde{r} = 1 + \beta t + v$. We temporarily omit the tildes.

Set $s = 1 + \beta t + v$ and write uniquely

$$p = A(t)s + B(t), \quad q = C(t)s + D(t), \quad r = s. \quad (5.2)$$

Substitution into (2.4) gives

$$\mathcal{J} = E_0 + E_1 s + E_2 s^2, \quad (5.3)$$

where the expressions E_0, E_1, E_2 are the left sides of (4.4)–(4.6). For clarity, differentiation at fixed v gives $p_t = A's + A\beta + B'$, $q_t = C's + C\beta + D'$, and $r_t = \beta$. Expanding the determinant cancels the β terms and yields (5.3). Since (t, s) are independent polynomial coordinates, the Keller equation is exactly (4.4)–(4.6).

The boundary conditions become

$$\begin{aligned} A(0) &= 1, & B(0) &= -1, & A'(0) + \beta + B'(0) &= 0, \\ C(0) + D(0) &= 0, & C'(0) + \beta C(0) + D'(0) &= 1. \end{aligned} \quad (5.4)$$

We must not divide by C before excluding $C = 0$. If C were zero, the fourth condition would give $D(0) = 0$. Equation (4.4) at zero would then give $D'(0) = -1/2$, while (4.5) at zero would give $-3D'(0) = 0$. This is impossible.

The rigidity lemmas now apply. Write

$$\begin{aligned} E &= 1 + kt, & A &= E^3, & C &= cE^2, & c &\neq 0, \\ D &= dt - c, & B &= -1 + b_1 t + b_2 t^2. \end{aligned} \quad (5.5)$$

The first derivative condition in (5.4) gives

$$b_1 = -\beta - 3k. \quad (5.6)$$

After dividing (4.5) by the nonzero polynomial E , it reads

$$2c(EB' - 2kB) = 3(kc + d)E. \quad (5.7)$$

Its constant and linear coefficients, together with (5.6), give

$$b_2 = -\beta k - 2k^2, \quad \frac{d}{c} = -\frac{2\beta + 5k}{3}. \quad (5.8)$$

The final condition in (5.4) is $c(\beta + 2k) + d = 1$, which therefore simplifies to

$$c(\beta + k) = 3. \quad (5.9)$$

The coefficient of t in (4.4) is $-db_1 - 2cb_2$. Its vanishing, using (5.6) and (5.8), is

$$\frac{c}{3}(\beta + k)(2\beta + 3k) = 0.$$

Equation (5.9) permits cancellation of the first factors. Hence

$$k = -\frac{2\beta}{3}, \quad \beta \neq 0, \quad c = \frac{9}{\beta}, \quad d = 4, \quad b_1 = \beta, \quad b_2 = -\frac{2\beta^2}{9}. \quad (5.10)$$

The remaining constant coefficient of (4.4) is indeed -1 : $-cb_1 + 2d = -9 + 8 = -1$. All three equations hold for these values. We have proved uniqueness and existence, including the exclusion of $\beta = 0$.

More compactly, before undoing (5.1), the answer is

$$\begin{aligned} E &= 1 - \frac{2\beta}{3}t, & s &= 1 + \beta t + v, \\ p &= E^3 s - 1 + \beta t - \frac{2\beta^2}{9}t^2, & q &= \frac{9}{\beta}E^2 s + 4t - \frac{9}{\beta}. \end{aligned} \quad (5.11)$$

Replace v by γv and divide p by γ . Expansion yields (3.1) with $a = \beta^2/\gamma$ and $b = \gamma/\beta$. This proves the nonconstant branch whenever $\gamma \neq 0$.

5.2 The case $\gamma = 0$, $\beta \neq 0$: no solutions

Write $p = A(t)v + B(t)$, $q = C(t)v + D(t)$, and $r = 1 + \beta t$. The normalization implies $A(0) = 1$ and $B(0) = B'(0) = D(0) = 0$. The coefficient of v in (2.4) is

$$\beta AC + r(A'C - AC') = 0. \quad (5.12)$$

If $C \neq 0$, this says that $(rA/C)' = 0$, so $C = \lambda rA$ for some $\lambda \in K^\times$. When $C = 0$, the same expression holds with $\lambda = 0$. The constant coefficient of the determinant then becomes

$$A(\lambda(r^2 B)' - (rD)') = -1. \quad (5.13)$$

A product of two polynomials is a nonzero constant only when both are constants. Thus $A = 1$. Integrating (5.13) and using $B(0) = D(0) = 0$ gives

$$\lambda r^2 B - rD = -t. \quad (5.14)$$

Evaluate at $t = -1/\beta$. The left side is zero and the right side is $1/\beta \neq 0$. This contradiction excludes the entire case, at every degree.

5.3 The case $r = 1$: the full tame branch

Again write $p = Av + B$ and $q = Cv + D$. The coefficient of v in the determinant is $A'C - AC' = 0$. Since $A(0) = 1$, it follows that $C = \lambda A$ for some $\lambda \in K$, allowing $C = 0$. The constant coefficient is

$$A(\lambda B' - D') = -1.$$

Consequently $A = 1$ and $D = t + \lambda B$, using $B(0) = D(0) = 0$. Polynomiality and normalization give $B(0) = B'(0) = 0$, hence $B \in t^2K[t]$. This is precisely (3.2). Conversely, these formulas give $\mathcal{J} = -1$ directly. Together with the preceding two cases, this completes the algebraic classification in Theorem 3.1.

6 Invertibility, diagonal equivalence, and sharp support

6.1 An inverse for every constant-third-numerator map

Write $B(t) = t^2H(t)$. The lift of (3.2) is

$$F(x, y, z) = (P, y + \lambda xP, x), \quad P = z + y^2H(xy). \quad (6.1)$$

For output coordinates (U, V, W) , define

$$x = W, \quad y = V - \lambda WU, \quad z = U - (V - \lambda WU)^2H(W(V - \lambda WU)). \quad (6.2)$$

Substitution in both directions is immediate from $P = U$. This proves polynomial invertibility. More precisely, (6.1) is the composition of the elementary shear $z \mapsto z + y^2H(xy)$, the elementary shear $y \mapsto y + \lambda xz$ in the updated coordinates, and the coordinate permutation $(x, y, z) \mapsto (z, y, x)$. It is therefore tame. There is no bound on $\deg H$: arbitrarily large degrees occur on the invertible branch.

6.2 Every nonconstant map is in the same diagonal orbit

For $a, b \neq 0$, inspection of (3.1) gives

$$F_{a,b} = \text{diag}((ab^2)^{-1}, (ab)^{-1}, 1) \circ F_* \circ \text{diag}(1, ab, ab^2). \quad (6.3)$$

The product of the two diagonal determinants is one, consistently with $\det J_{F_{a,b}} = -1$. Thus there are two parameters before taking this source–target equivalence, and no parameter after taking it. The scalar parameters are uniquely recoverable from either r or two coefficients:

$$a = -\frac{9}{8}[t^2]p, \quad b = \frac{1}{9}[v]q. \quad (6.4)$$

Corollary 6.1 (Sharp all-degree support rigidity). *Within the class of Theorem 3.1, every noninvertible map has ordinary maximum degree seven and exactly sixteen expanded monomials. After ordering coordinates by their prescribed weights, its degree profile is $(7, 6, 4)$ and its support profile is $(7, 6, 3)$.*

Proof. Only branch (B) can be noninvertible. For $a, b \neq 0$, all coefficients displayed in (3.1) are nonzero. Different monomials in a numerator lift to different monomials. The degree of a lifted $t^i v^j$ is $2i + 3j - 2$ in the first coordinate, $2i + 3j - 1$ in the second, and $2i + 3j + 1$ in the third. The asserted profiles follow. Noninjectivity of each map is established just below. $\square$

This is stronger than saying that sixteen terms are minimal under a degree cap: every noninvertible map in the structural class has that precise support profile, regardless of the degree initially allowed. It is not a statement about minimum degree or support outside the class.

6.3 Collisions valid over the ground field

Let $\gamma = ab^2$ and $\ell = 9b$. Direct evaluation on the plane $x = 0$ gives

$$F_{a,b}(0, y, z) = \left(z - \frac{8a}{9}y^2, y, 0 \right). \quad (6.5)$$

At $t = 0$, the numerators simplify to $p = v$, $q = \ell v$, and $r = 1 + \gamma v$. Therefore

$$F_{a,b} \left(1, 0, -\frac{1}{\gamma} \right) = F_{a,b} \left(0, -\frac{\ell}{\gamma}, \frac{71}{\gamma} \right) = \left(-\frac{1}{\gamma}, -\frac{\ell}{\gamma}, 0 \right). \quad (6.6)$$

To check the constant 71, note that $-(8a/9)\ell^2 = -72\gamma$; the first coordinate of the second point's image is therefore $(71 - 72)/\gamma$.

The two source points have different x coordinates. This proves noninjectivity over *every* characteristic-zero ground field K , without passing to an algebraic closure. It also gives a rational certificate when $a, b \in \mathbb{Q}^\times$.

Equivariance propagates this to every $\rho \in K^\times$:

$$\begin{aligned} F_{a,b} \left(\rho, 0, -\frac{1}{\gamma\rho^2} \right) &= F_{a,b} \left(0, -\frac{\ell}{\gamma\rho}, \frac{71}{\gamma\rho^2} \right) \\ &= \left(-\frac{1}{\gamma\rho^2}, -\frac{\ell}{\gamma\rho}, 0 \right). \end{aligned} \quad (6.7)$$

These are explicit certificates in the coordinates of this report. They are consequences of the classified family, not a claim that the collision mechanism of the original map was unknown. [write] The mechanism is the project's (`Algebra/JacobianConjecture/Research/README.md`, lines 210–224): (6.6) is its case $t_0 = 0$, $v_0 = -1/\gamma$, and the second point $(0, q, p - \alpha q^2)$ with $\alpha = [t^2]p_{a,b} = -8a/9$ (the notes' a) is the one displayed. For $a = b = 1$ this was checked at the write by exact evaluation.

Example 6.2 (The normalized rational witness). For $a = b = 1$, the points $(1, 0, -1)$ and $(0, -9, 71)$ have common image $(-1, -9, 0)$. The Jacobian is nonsingular at both points because it has constant determinant -1 . Thus local nonsingularity and global injectivity are genuinely different conditions in the displayed family.

6.4 A larger class: arbitrary dependence on the first invariant

The affine restriction on r can be relaxed substantially without changing its essential normal form. The relevant source automorphisms are

$$T_h(x, y, z) = (x, y, z + y^2 h(xy)), \quad h \in K[t]. \quad (6.8)$$

They preserve the weights, have Jacobian one, and induce $(t, v) \mapsto (t, v + t^2 h(t))$. Their inverses are T_{-h} .

Theorem 6.3 (All-degree triangular extension). *Let p, q be affine in v and let*

$$r = R(t) + \gamma v, \quad R \in K[t], \quad \gamma \in K,$$

with polynomial lift and normalized origin coefficients. In particular, $R(0) = 1$. Then the normalized lift is Keller if and only if one of the following holds.

- (i) $\gamma = 0$, $R = 1$, and the map is a tame map (6.1).
- (ii) $\gamma \neq 0$, $\beta := R'(0) \neq 0$, and the map is

$$F = F_{a,b} \circ T_h, \quad a = \frac{\beta^2}{\gamma}, \quad b = \frac{\gamma}{\beta}, \quad h(t) = \frac{R(t) - 1 - \beta t}{\gamma t^2}. \quad (6.9)$$

Here h is a polynomial, uniquely determined by r . The map is noninjective over K .

Thus every noninvertible map in this larger class is the classified representative up to diagonal scalings and one explicit equivariant source shear. There is no degree bound on R , p , or q .

Proof. When $\gamma \neq 0$, the numerator defining h vanishes to order at least two at zero. Composing the source with T_{-h} replaces v by $v - t^2 h(t)$ and changes the third numerator to $1 + \beta t + \gamma v$. The first two numerators remain affine in v ; polynomiality and all normalized origin coefficients are preserved. The source shear has Jacobian one. Theorem 3.1 therefore applies: it forces $\beta \neq 0$ and identifies $F \circ T_{-h}$ with $F_{a,b}$. This is (6.9). The converse follows from the same composition, and collision points are transported by T_{-h} .

When $\gamma = 0$, use the proof of Section 5 with $r = R(t)$ instead of $1 + \beta t$. The coefficient of v is $r'AC + r(A'C - AC') = 0$, whence $C = \lambda rA$, including $C = 0$. The constant equation is still $A(\lambda(r^2B)' - (rD)') = -1$. It forces $A = 1$ and, on integration with the origin conditions, $\lambda r^2B - rD = -t$. Thus r divides t . Since $r(0) = 1$, this forces r to be constant and hence $r = 1$. The already proved tame classification finishes this case. $\square$

Corollary 6.4 (An exact degree spectrum). *In the noninvertible branch of Theorem 6.3, $h = 0$ gives coordinate degrees $(7, 6, 4)$. If $h \neq 0$ has degree m , the coordinate degrees are exactly*

$$(2m + 8, 2m + 7, 2m + 5). \quad (6.10)$$

Consequently the possible maximum ordinary degrees in this noninvertible class are precisely 7 and every even integer at least 8.

Proof. The coefficients of z in the three coordinates of $F_{a,b}$ have ordinary degrees 6, 5, 3, respectively. Their leading terms are nonzero for $a, b \neq 0$. The substitution $z \mapsto z + y^2 h(xy)$ adds terms of degrees $6 + (2 + 2m)$, $5 + (2 + 2m)$, and $3 + (2 + 2m)$. These are all strictly larger than the original coordinate degrees, so they cannot cancel against an original term. Their leading coefficients are nonzero, proving (6.10). Every $m \geq 0$ occurs by taking $h = t^m$. $\square$

The first invariant coefficient is the important distinction here. The coefficient of v in r is still required to be a scalar, not a nonconstant polynomial in t . No assertion is made for $r = R(t) + G(t)v$ with nonconstant G .

7 Reduced parameter geometry and invertible degenerations

The formulas (3.1) contain no negative powers of a or b . They therefore define a polynomial family on the entire parameter plane, not just on its torus. The diagonal formula (6.3) should not be used on the axes, where its individual scaling matrices are undefined. The polynomial formulas are the correct extension there.

Theorem 7.1 (The parameter plane and its boundary). *In the finite coefficient space of normalized weighted maps of ordinary degree at most seven with p, q affine in v and r affine in (t, v) , the following statements hold over K .*

- (i) The morphism $(a, b) \mapsto F_{a,b}$ is a closed embedding of $\mathbb{A}_K^2$. Every member has Jacobian determinant -1 .
- (ii) Its torus $ab \neq 0$ is exactly the normalized nonconstant- r branch. Its reduced Zariski closure is the entire embedded affine plane.
- (iii) The invertible members of this plane are exactly its two coordinate axes $V(ab)$. Explicitly,

$$F_{0,b}(x, y, z) = (z, y + 9bxz, x), \quad (7.1)$$

$$F_{a,0}(x, y, z) = \left(z - \frac{8a}{9}y^2, y, x \right). \quad (7.2)$$

- (iv) Inside this plane, the scheme-theoretic condition $r = 1$ cuts out the reduced nodal union $\text{Spec } K[a, b]/(ab)$.

Proof. The coefficients (6.4) recover a, b by linear functions on the ambient coefficient space. Every other coefficient is a polynomial in those two functions by (3.1). Thus the image is the graph of polynomial functions, a closed embedding.

The determinant is -1 on the torus by the classification and its explicit solution. It is a polynomial in a, b, t, v , so the identity holds on the whole plane. Equivalently, it can be checked directly from (3.1). The torus is dense, and the classification identifies all its field-valued nonconstant- r points. This proves the closure statement.

On the axes, the formulas reduce to (7.1) and (7.2), which are tame maps of the already classified constant branch. Off the axes, (6.6) excludes invertibility. Finally, imposing $r = 1$ on the parameter ring imposes $(ab, ab^2) = (ab)$. The ideal (ab) is radical: it is the intersection $(a) \cap (b)$ in $K[a, b]$. $\square$

7.1 What degenerates, and what does not

The Jacobian determinant stays -1 throughout this degeneration. The ordinary degrees and the collision coordinates do not stay bounded in the same manner. Fixing $b = 1$ and letting a approach zero over $\mathbb{R}$ or $\mathbb{C}$, the coefficients tend to the automorphism $F_{0,1}$, whereas the two source points in (6.6) have coordinates with poles in a . In this explicit sense the displayed collisions escape to infinity. No bounded limiting pair of distinct colliding points is asserted.

There are many other constant- r maps, because $B \in t^2 K[t]$ in (3.2) is arbitrary. Only the two axes described above lie in the closure of the nonconstant branch in this coefficient space. In particular, passing from the nonconstant branch to its closure does not recover the entire infinite-dimensional tame family.

7.2 Rigidity over every reduced coefficient algebra

Field-valued uniqueness can be strengthened without any degree cap on the numerators.

Corollary 7.2 (All-degree reduced rigidity). *Let R be a reduced $\mathbb{Q}$ -algebra. Suppose $p, q \in R[t, v]$ are affine in v , satisfy the normalization and polynomiality conditions, and satisfy $\mathcal{J}(p, q, 1 + t + v) = -1$. Then $p = p_*$ and $q = q_*$.*

Proof. For every prime ideal $\mathfrak{p}$ of R , pass to the fraction field of $R/\mathfrak{p}$. It is a characteristic-zero field, so Theorem 3.1 forces $a = b = 1$. Every coefficient of $p - p_*$ and $q - q_*$ therefore belongs to every prime ideal of R , hence to its nilradical. That nilradical is zero because R is reduced. $\square$

Thus fixing $r = 1 + t + v$ eliminates nonconstant families over reduced bases, even when large ordinary degrees are allowed. It does *not* eliminate families over nonreduced bases. We now determine one such coefficient scheme exactly.

8 The normalized coefficient scheme is a double point

The preceding proof used degree arguments over fields. Nilpotent leading coefficients need not satisfy those arguments. Scheme structure must therefore be analyzed rather than inferred from the unique field-valued solution.

8.1 The precise finite coefficient problem

Fix $r = 1 + t + v$ and ordinary degree at most seven. The most general normalized numerators in our class are

$$p = v + p_{20}t^2 + p_{11}tv + p_{30}t^3 + p_{21}t^2v + p_{40}t^4 + p_{31}t^3v, \quad (8.1)$$

$$q = t + q_{01}v + q_{20}t^2 + q_{11}tv + q_{30}t^3 + q_{21}t^2v + q_{40}t^4. \quad (8.2)$$

The degree formulas in the proof of Corollary 6.1 show that this list is exhaustive. In particular, the q_{40} term is allowed and must not be set to zero beforehand.

Let the coefficient vector, in the following fixed order, be

$$c = (p_{20}, p_{11}, p_{30}, p_{21}, p_{40}, p_{31}, q_{01}, q_{20}, q_{11}, q_{30}, q_{21}, q_{40}). \quad (8.3)$$

Let $I \subset \mathbb{Q}[c]$ be generated by all coefficients in t, v of $\mathcal{J}(p, q, 1 + t + v) + 1$. There are eighteen nonzero coefficient polynomials. Their complete expressions, ordering, and derivative matrix are supplied in `verification.json`; the article fixes enough data below to reconstruct and check the certificates independently.

Theorem 8.1 (Exact double-point presentation). *The full coordinate algebra of this coefficient problem is*

$$\mathbb{Q}[c]/I \simeq \mathbb{Q}[\varepsilon]/(\varepsilon^2). \quad (8.4)$$

Under this isomorphism $\varepsilon = q_{21} - 4$ and

$$c = c_* + u\varepsilon, \quad (8.5)$$

where

$$c_* = \left(-\frac{8}{9}, -2, \frac{28}{27}, \frac{4}{3}, -\frac{8}{27}, -\frac{8}{27}, 9, -8, -12, 4, 4, 0 \right) \quad (8.6)$$

and

$$u = \frac{1}{144}(-7, -18, 18, 24, -8, -8, 162, -198, -324, 144, 144, 0). \quad (8.7)$$

In particular, the entire ideal, not merely its radical, is generated by ε^2 and the eleven linear relations obtained from (8.5) by omitting the tautological q_{21} relation.

The proof has three components: uniqueness of geometric points, a rank certificate for the tangent space, and a certificate excluding a second-order lift. None requires a Gröbner-basis computation.

8.2 Tangent space and its exact rank certificate

Write the coefficient equations as $f_1, \dots, f_{18}$ in the order of the monomials

$$\begin{aligned} t^7, t^6v, t^6, t^5v, t^5, t^4v, t^4, t^3v^2, t^3v, t^3, \\ t^2v^2, t^2v, t^2, tv^2, tv, t, v^2, v. \end{aligned} \quad (8.8)$$

Let

$$M = \left(\frac{\partial f_i}{\partial c_j}(c_*) \right)_{i,j}. \quad (8.9)$$

Direct rational calculation gives $Mu = 0$. Delete the column corresponding to q_{21} and select the rows corresponding to

$$t^7, t^6, t^5, t^4v, t^4, t^3, t^2v, t^2, tv^2, t, v. \quad (8.10)$$

The determinant of that 11×11 submatrix is

$$-18\,874\,368 \neq 0. \quad (8.11)$$

Thus $\text{rank } M = 11$ and $\ker M = \mathbb{Q}u$. This proves that the scheme has a one-dimensional Zariski tangent space at its sole geometric point. It does not yet determine the length of the point.

Because $f_i(c_*) = 0$ and $Mu = 0$, the assignment $c = c_* + u\varepsilon$ solves the equations over $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$. Since $u_{q_{21}} = 1$, the resulting map from the coefficient algebra onto the dual numbers is surjective. In particular, the tangent direction is not merely a redundant parametrization of a reduced point.

8.3 A nonzero obstruction at second order

Define the quadratic residual vector

$$H = ([\varepsilon^2]f_i(c_* + u\varepsilon))_{i=1}^{18}. \quad (8.12)$$

Consider a possible lift to $\mathbb{Q}[\varepsilon]/(\varepsilon^3)$ of the form $c = c_* + u\varepsilon + w\varepsilon^2$. Its second-order equation is

$$Mw + H = 0. \quad (8.13)$$

There is an explicit row functional

$$L = (-205, 0, -46, 0, -40, 24, -8, 0, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0) \quad (8.14)$$

satisfying

$$LM = 0, \quad LH = -\frac{1}{72}. \quad (8.15)$$

Applying L to (8.13) produces a contradiction. For a nonzero scalar multiple ηu , the obstruction pairing is $-\eta^2/72$, which is again nonzero over every characteristic-zero field. There is therefore no second-order lift of any nonzero first-order tangent direction. The matrix, vector, and pairing in (8.9)–(8.15) are checked exactly by the included script.

8.4 From tangent obstruction to the entire scheme

We record the small algebraic fact that turns these certificates into a scheme-theoretic theorem.

Lemma 8.2 (An isolated point with one obstructed tangent). *Let k be algebraically closed and let A be a finitely generated k -algebra with exactly one geometric point and residue field k . Suppose its tangent space has dimension one and no nonzero tangent homomorphism $A \rightarrow k[\varepsilon]/(\varepsilon^2)$ extends to $A \rightarrow k[\varepsilon]/(\varepsilon^3)$. Then $A \simeq k[z]/(z^2)$.*

Proof. The Nullstellensatz implies that the radical of zero is the unique maximal ideal $\mathfrak{m}$ [6]. The algebra is Noetherian. A finitely generated ideal consisting of nilpotent elements is nilpotent: if its generators have nilpotence exponents n_i , every product of more than $\sum_i (n_i - 1)$ generators is zero. Thus some power of $\mathfrak{m}$ is zero, and A is a finite-dimensional local k -algebra.

The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ has dimension one. Choose $z \in \mathfrak{m}$ whose residue spans it. Nakayama's lemma gives $\mathfrak{m} = (z)$ [7]. Repeatedly writing an element as a scalar plus z times another element, and using nilpotence, shows that A is generated by z over k . Since $k[Z]$ is a principal ideal domain and the only root of the defining ideal is zero, it follows that $A \simeq k[Z]/(Z^N)$ for some N . The nonzero cotangent space forces $N \geq 2$. If $N \geq 3$, sending Z to ε defines a homomorphism to $k[\varepsilon]/(\varepsilon^3)$ with nonzero first-order part, contrary to the hypothesis. Hence $N = 2$. $\square$

Proof of Theorem 8.1. Work first over $\overline{\mathbb{Q}}$. The classification with $\beta = \gamma = 1$ forces $a = b = 1$, so the coefficient scheme has exactly one geometric point, c_* . Equations (8.11) and $Mu = 0$ give tangent dimension one, and (8.15) excludes every nonzero second-order lift. Lemma 8.2 gives a two-dimensional coordinate algebra over $\overline{\mathbb{Q}}$.

Already over $\mathbb{Q}$, substitution $c = c_* + u\varepsilon$ defines a surjection

$$\mathbb{Q}[c]/I \longrightarrow \mathbb{Q}[\varepsilon]/(\varepsilon^2).$$

After extension to $\overline{\mathbb{Q}}$, this is a surjection between vector spaces of dimension two and is an isomorphism. Field extension is faithful, so its kernel over $\mathbb{Q}$ was zero. This proves (8.4) with the specified coordinates.

The kernel of the explicit substitution map from the polynomial ring $\mathbb{Q}[c]$ is the ideal generated by ε^2 and the eleven affine-linear relations $c_j - (c_*)_j - u_j\varepsilon$. The isomorphism identifies this kernel with I , proving the exact ideal assertion. $\square$

8.5 The universal meaning of the answer

Corollary 8.3 (Solutions over arbitrary coefficient algebras). *For every $\mathbb{Q}$ -algebra R , the solutions of (8.1)–(8.2) with $\mathcal{J} = -1$ are in bijection with*

$$\{e \in R : e^2 = 0\}.$$

The corresponding solution is $c = c_ + ue$, and $e = q_{21} - 4$.*

Proof. Apply $\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(-, R)$ to (8.4). $\square$

In particular, every solution over a reduced ring is the constant one, but a coefficient ring containing a nonzero square-zero element supports a nontrivial infinitesimal deformation. The parameter is obstructed rather than the initial segment of a moving curve of normalized field-valued solutions. The square seen in the repository's finite coefficient certificate is therefore genuine scheme structure, not just an artifact of one Gröbner-basis presentation.

8.6 The variable-parameter open scheme

The nonconstant coefficient parameters can also be restored without losing control of nilpotents. Let $\mathcal{X}^\times$ denote the coefficient scheme obtained by using (8.1)–(8.2), putting $r = 1 + \beta t + \gamma v$, and inverting $\beta\gamma$.

Theorem 8.4 (Torus times the double point). *There is an explicit isomorphism*

$$\mathcal{X}^\times \simeq \operatorname{Spec} \mathbb{Q}[\beta^{\pm 1}, \gamma^{\pm 1}, \varepsilon]/(\varepsilon^2). \quad (8.16)$$

It is an isomorphism of full schemes, not just of reduced varieties. The square-zero coordinate can be taken to be $\varepsilon = q_{21}/(\beta\gamma) - 4$.

Proof. Over any $\mathbb{Q}$ -algebra where β, γ are units, define

$$\bar{p}(t, v) = \gamma p(t/\beta, v/\gamma), \quad \bar{q}(t, v) = \beta q(t/\beta, v/\gamma). \quad (8.17)$$

The third numerator becomes $1 + t + v$. The two normalized origin coefficients remain one. These substitutions preserve the allowed monomial supports and ordinary degree bound, and have polynomial inverse formulas over the Laurent coefficient ring. They preserve the determinant condition: the row scalings contribute $\beta\gamma$ and the derivative column scalings contribute $(\beta\gamma)^{-1}$.

Thus (8.17) identifies the coefficient functor with a choice of two units and a solution of the fixed normalized scheme. Apply Theorem 8.1 to that fixed scheme. The coefficient of t^2v in $\bar{q}$ is $q_{21}/(\beta\gamma)$, giving the stated square-zero coordinate. All formulas are valid with nilpotent coefficients, so this proves the scheme isomorphism. $\square$

At every geometric point of this open scheme the reduced space has dimension two and the tangent space has dimension three: two tangent directions come from the torus and one from its square-zero thickening. The remaining boundary problem is how this structure meets the axes, where the Laurent normalization is no longer available.

Remark 8.5 (The degree boundary matters). Corollary 7.2 is an all-degree statement. In contrast, Theorem 8.1 fixes the ordinary degree-seven coefficient space. We have not proved that allowing higher-degree coefficients over nonreduced rings introduces no extra tangent directions or higher-order nilpotents. The field-valued classification alone does not imply that stronger assertion.

9 Decision procedure, reproducibility, and formalization

9.1 A classification-based recognition procedure

The theorem gives more than a determinant identity. For rational input polynomials in the structural class, it yields a direct recognition procedure for the Keller condition, together with an inverse or collision certificate.

First check the prescribed weights monomial by monomial and extract the unique invariant numerators as in Section 2. Check $\deg_v p, \deg_v q \leq 1$ and that r is affine. Inputs outside these restrictions are *outside the procedure's scope*; they are not rejected as non-Keller maps.

Within this scope, compute $p_v(0), q_t(0), r(0)$. If one is zero, the map is not Keller by (2.5). Otherwise normalize the target coordinates. Write $r = 1 + \beta t + \gamma v$. If $\beta = \gamma = 0$, take $B = p - v$ and $\lambda = [v]q$, and test the exact identities (3.2) with $B \in t^2K[t]$. If exactly one of β, γ is nonzero, the map is not Keller. If both are nonzero, recover a, b from (3.3) and compare

with the fixed formula (3.1). The classification proves both soundness and completeness of these comparisons inside the stated scope.

For a positive result, return either the inverse (6.2) or the pair of source points in (6.6). To undo target normalization, compose an inverse with the inverse target scaling; collision source points remain collision source points, with their common target rescaled. This avoids a search over all possible inverses or collision pairs.

The larger class of Theorem 6.3 is recognized by first computing its unique source shear. For $\gamma \neq 0$, compute h from (6.9), compose the source with T_{-h} , and apply the affine- r procedure. For $\gamma = 0$, a nonconstant $R(t)$ is excluded by that theorem. Inputs with nonconstant v -coefficient in r remain outside the classified scope.

For canonical sparse input, reading and comparing the coefficient lists requires a number of field operations linear in the input support size, apart from the cost of canonicalizing the input. This is an arithmetic operation statement, not a claim of linear bit complexity for arbitrarily large rational coefficients. An implementation should retain that distinction.

9.2 What the companion script checks

The file `verify_results.py` ([write] shipped as `code/verify_results.py`, with its record in `data/`) works in exact rational and symbolic arithmetic. It reconstructs the equations rather than trusting a stored matrix. Its thirteen groups of checks cover:

1. The two-parameter determinant in both invariant coordinates and the full three-variable Jacobian; the original-map normalization and collision; diagonal equivalence, degrees, support, and the universal collision family.
2. The two degenerate axes, the two-sided inverse identity with a formal arbitrary function H , the source-shear identities and sample degree profiles, and the one-variable coefficient identities and explicit endpoint of the classification proof.
3. The invertible variable-parameter scheme normalization, the eighteen coefficient equations, the rank minor, the tangent vector, the cokernel functional, the second-order obstruction, and the vanishing of every coefficient equation under the dual-number presentation.

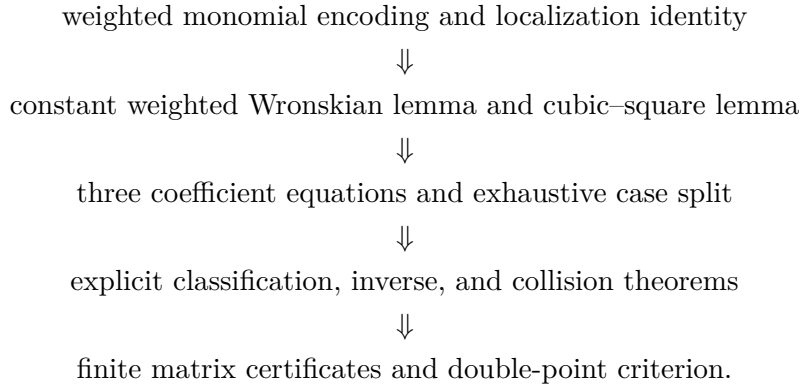
The last check proves one ideal containment by direct substitution. The reverse containment is justified in the article by the classification, rank, and obstruction argument; the script does not silently label a one-sided ideal check as equality.

The run included with this report used SymPy 1.14.0 and passed all thirteen check groups. The JSON certificate includes the full rational matrix and both row and column orderings. No numerical tolerance, random test, or floating-point approximation is used. These checks are reproducibility evidence for the finite algebra; they do not replace the universal one-variable degree proofs.

9.3 A realistic proof-assistant boundary

The new theorems in this report have not been compiled in Lean or Rocq. The existing repository formalizations of the original map are separate from the new classification. A

sensible formalization order is the following dependency chain:



The main classification needs polynomial degrees, characteristic-zero derivatives, a rational-function constant lemma, and unique factorization in one variable. It does not require the heavy geometric machinery used in the literature to study unrestricted Keller maps. The double-point statement additionally needs finite-type algebra, the Nullstellensatz, Nakayama's lemma, and faithful scalar extension, or an alternative direct ideal-membership certificate.

One attractive alternative for formalizing the last step is to produce explicit polynomial combinations expressing the twelve simple generators of the double-point ideal in terms of the eighteen original equations. The present argument proves these combinations exist but does not supply their expanded coefficients. Such certificates could isolate the formal scheme result from a large commutative-algebra library.

10 Further research questions and proposed next steps

The following questions are proposed by the present results. Unless explicitly attributed, they are research directions formulated here, not claims that a published source listed them as open. They are ordered to separate immediately testable extensions from much broader classification problems.

10.1 Allow quadratic dependence on the second invariant

Keep r affine but allow $\deg_v p, \deg_v q \leq 2$. Does the nonconstant branch admit genuinely new diagonal-equivalence classes? After the change $s = r$, the determinant becomes a polynomial of higher degree in s . The highest coefficient should impose a weighted differential relation on the leading coefficients, but the lower equations can couple several common factors. The first concrete task is to derive the complete coefficient system and identify whether an analogue of Lemma 4.1 gives an a priori degree bound. A finite search would be useful evidence, but an all-degree conclusion needs such a bound or another structural argument.

10.2 Allow a nonconstant second-invariant coefficient in r

Theorem 6.3 settles arbitrary polynomial t -dependence when the coefficient of v in r is constant. The next case is $r = R(t) + G(t)v$ with nonconstant $G(t)$. Must G be constant under suitable Keller and polynomiality hypotheses, or do genuinely new normal forms appear? The substitution $s = r$ now requires division by G and is not a global polynomial coordinate change. Track the behavior at the zeros of G and determine whether the constant-Wronskian argument has a localized analogue. Allowing a quadratic term in v is a further, distinct extension.

10.3 Does the nonreduced structure stabilize with degree?

For each $d \geq 7$, form the normalized coefficient scheme with $r = 1 + t + v$, p, q affine in v , and lifted ordinary degree at most d . Its reduced scheme is a point by Corollary 7.2. Is the full scheme always the same double point? Or can high-degree coefficients support new nilpotent directions invisible at degree seven? A first step is to linearize the three identities in (A, B, C, D) around the explicit solution without truncating their degrees, solve the resulting polynomial linear differential system, and then examine its obstruction maps. This is a particularly well-targeted extension of the present work.

10.4 Scheme-theoretic gluing at the invertible boundary

Theorem 8.4 settles the full degree-seven coefficient scheme on $\beta\gamma \neq 0$: it is the parameter torus times a double point. Determine how this thickening meets the constant- r tame locus where that normalization ceases to be invertible. Does its scheme-theoretic closure acquire embedded components or higher nilpotent order on either axis or at their intersection? The reduced affine-plane closure and the nodal boundary in Theorem 7.1 do not decide this gluing problem. A concrete first step is to saturate the coefficient ideal by $\beta\gamma$, then study its specialization on each axis with explicit ideal-membership certificates.

10.5 Other weights and other Wronskian exponents

Our determinant produces a cubic–square relation because the first two weights are 2 and 1. For other mixed-sign weight triples, analogous highest-coefficient equations may have the form $mCA' - nAC' = 0$. Investigate the resulting common-power factorizations and constant weighted Wronskians. The interesting issue is not just repeating the calculation with new constants: the polynomiality inequalities at the coordinate hyperplane may either permit or rule out the required factorizations. A systematic analysis could delineate entire rigid classes of equivariant maps.

10.6 Compare lower-degree maps outside this symmetry class

The current literature already includes a three-dimensional degree-four example [5]; the existence of such an example is therefore not posed as a new open problem here. ([write] Gao’s degree is the geometric degree; the ordinary degrees of his map are 4, 11, 12, see Section 1.5. A three-variable Keller map of *ordinary* degree below seven is not known from the sources cited here.) Instead, extract its invariant or geometric structure and determine precisely which hypothesis of our classification fails. Compare support size, rational coefficient height, and the cost of a formal collision certificate. This would help replace a single “simplest map” question by explicit, noninterchangeable complexity criteria.

10.7 Arithmetic optimization within the diagonal orbit

Over $\mathbb{Q}$, the classification makes every nonconstant map in this class explicitly diagonal-equivalent. One can now optimize primitive coefficient height, denominator size, or integral collision height within that orbit. The parameters a, b and the additional unnormalizing target scalings make these concrete arithmetic optimization problems. Prove lower bounds for a specified normalization rather than relying on finite samples of small rational parameters.

10.8 Direct ideal certificates and kernel verification

Generate explicit membership certificates for the twelve simple generators in Appendix A, and check them with polynomial normalization in Lean or Rocq. This would turn the deformation theorem into a compact finite certificate after the all-degree classification has been formalized. The target should be a transparent theorem boundary: no hidden use of a computer-algebra claim that a Gröbner basis is complete.

10.9 Positive characteristic as a controlled contrast

The displayed family and finite identities have meaningful reductions at many primes, but the all-degree proof does not automatically survive. In positive characteristic, nonconstant p th powers have derivative zero, and the weighted degree coefficients can vanish. Classify which parts of the theorem persist under explicit degree bounds and which fail through Frobenius phenomena. Separately determine the primes at which the tangent rank or obstruction changes. Reducing a characteristic-zero certificate is not sufficient to claim the same unrestricted classification.

11 Conclusion

The repository's finite sparsity problem admits a stronger structural answer in its affine-invariant class. Constant third numerator gives all and only the explicit tame maps (3.2). Nonconstant third numerator forces both invariant coefficients to be nonzero and forces the entire map into the diagonal orbit (3.1), with degree seven and sixteen terms. An explicit source shear extends the classification to $r = R(t) + \gamma v$ and gives the exact degree spectrum $\{7, 8, 10, 12, \dots\}$. The proof is an all-degree argument, not an extrapolation from a coefficient search.

The parameter-space results explain two different notions of rigidity. Before fixing the third numerator, the nonconstant branch has an affine plane closure with invertible coordinate axes. After fixing $r = 1 + t + v$, all reduced families are constant, yet the degree-seven coefficient scheme retains exactly one square-zero parameter. Its tangent direction has an explicit nonzero second-order obstruction. Restoring the two nonconstant parameters gives the torus times this double point. These distinctions are essential both for further searches and for a trustworthy formalization.

A An explicit presentation of the coefficient ideal

For readers who prefer generators to the vector formula (8.5), Theorem 8.1 says that I is exactly the ideal generated by

$$\begin{aligned}
 & q_{40}, \quad (q_{21} - 4)^2, \\
 & 144p_{20} + 7q_{21} + 100, \\
 & 8p_{11} + q_{21} + 12, \\
 & 216p_{30} - 27q_{21} - 116, \\
 & 6p_{21} - q_{21} - 4, \\
 & 54p_{40} + 3q_{21} + 4, \\
 & 54p_{31} + 3q_{21} + 4, \\
 & 8q_{01} - 9q_{21} - 36, \\
 & 8q_{20} + 11q_{21} + 20, \\
 & 4q_{11} + 9q_{21} + 12, \\
 & q_{30} - q_{21}.
 \end{aligned} \tag{A.1}$$

These equations apply over every $\mathbb{Q}$ -algebra, not just over fields. Over a field the square forces $q_{21} = 4$, and all remaining coefficients are the point (8.6). Over the dual numbers, $q_{21} = 4 + \varepsilon$ produces exactly the tangent vector (8.7).

As a useful independent check of the equation reconstruction, the coefficients corresponding to the last three monomials t, v^2, v in (8.8) are, respectively,

$$\begin{aligned}
 f_{16} &= -p_{11} + 2p_{20}q_{01} - 2q_{20} - 2, \\
 f_{17} &= 2p_{11}q_{01} - 3q_{11}, \\
 f_{18} &= p_{11}q_{01} + q_{01} - q_{11} - 3.
 \end{aligned}$$

For the first two monomials they are

$$f_1 = -5p_{31}q_{40} - 4p_{40}q_{40}, \quad f_2 = -9p_{31}q_{40}.$$

The remaining equations are recovered by the same determinant expansion; the complete list is included in the JSON certificate.

B Small exact certificates for the deformation calculation

The primitive integral tangent vector is $n = 144u$. To keep the second-order residual integral, define

$$\hat{H}_i = [e^2]f_i(c_* + en) = 144^2 H_i.$$

In the ordering (8.8), the nonzero data and the cokernel functional are as follows. Zeros are included to make the ordering unambiguous.

Row	Monomial	$\widehat{H}_i$	L_i
1	t^7	0	-205
2	t^6v	0	0
3	t^6	0	-46
4	t^5v	0	0
5	t^5	1008	-40
6	t^4v	1872	24
7	t^4	-4140	-8
8	t^3v^2	864	0
9	t^3v	-9504	4
10	t^3	2772	0
11	t^2v^2	-5184	0
12	t^2v	12636	0
13	t^2	1620	0
14	tv^2	9720	0
15	tv	-2592	0
16	t	-2268	0
17	v^2	-5832	0
18	v	-2916	0

The obstruction is visible in four integer products:

$$(-40)(1008) + 24(1872) + (-8)(-4140) + 4(-9504) = -288.$$

Dividing by 144^2 gives $LH = -1/72$.

For the nonzero minor in (8.11), the zero-based row indices in the JSON matrix are

$$(0, 2, 4, 5, 6, 9, 11, 12, 13, 15, 17),$$

and the zero-based column indices are

$$(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11).$$

Thus an independent checker only needs the reconstructed derivative matrix, one 11×11 determinant, and the two matrix-vector products Mu and LM . The companion JSON stores the matrix itself as rational strings for convenient cross-checking.

C Reproduction and source ledger

The supplied archive contains the following files:

File	Purpose
<code>article.tex</code>	Self-contained LaTeX source, including bibliography.
<code>article.pdf</code>	Compiled article.
<code>verify_results.py</code>	Reconstructs and checks all finite identities.
<code>verification.json</code>	Exact equations, matrix, and certificate data.
<code>verification.txt</code>	Output of the successful verification run.
<code>README.md</code>	Build commands, scope, and trust boundary.
<code>requirements.txt</code>	Reproducible SymPy dependency.

Build the article with two runs of `pdflatex article.tex`, or with `latexmk -pdf article.tex`. Run the checks with

```
python verify_results.py --output verification.json
```

[write] In ProveIt the script is `code/verify_results.py`; the records and `requirements.txt` are in `data/`; the delivered PDF is not shipped (`article.pdf` is a build of this text); the delivered README is replaced by the report README. The command above is the delivery's; its default output `verification.json` is relative to the working directory, so in ProveIt pass an output path outside the report (Appendix D.3).

The verification does not download or execute any repository code. The repository is a cited mathematical source; its original map and the relevant formulas are reconstructed in the script. The source ledger is:

Source category	Role in the report
Repository map	Original map, its known collision, and the finite-search baseline; not new claims of this article.
External research	Context for graded maps and examples outside our class; not dependencies of the classification proof.
Standard algebra	Nullstellensatz and Nakayama's lemma in the passage from an isolated obstructed tangent to a double point.
This article	All-degree classification, family closure, reduced rigidity, and the full degree-seven coefficient-algebra identification.

D Provenance of this report

D.1 One manuscript

[write] This report is built from *one* manuscript: manuscript 01 of batch 36 of ProveIt's incoming reports, the archive `ProveIt_Weighted_Keller_Rigidity` (inner directory of the same name, main file `article.tex`, a 23-page PDF dated 24 September 2026, PDF metadata author "ChatGPT; prepared for Vladimir Reshetnikov"), pinned to ProveIt commit `e21766d04c2b8a9b2cd43e563024b3bd1b9`. The archive arrived in commit `e13affd32` and was placed in `1a1396d4d` as a new report. Nothing was merged and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed. The manuscript contributed everything except the text marked [write].

The editorial changes are these.

- (1) Every label carries the prefix `wkr:`. The source's 96 labels are kept, unchanged after the prefix. Six were added: `wkr:sub:project`, `wkr:sub:gao`, `wkr:app:report`, `wkr:app:onesource`, `wkr:app:pinned` and `wkr:app:rerun`.
- (2) Sections 1.4 and 1.5 are new and end Section 1, so no source number moved. They credit the project's weighted determinant identity and collision mechanism, state the project's formal status, and correct the reading of Gao's "degree four".
- (3) Notes marked [write] were added on the title page, after the proof of Proposition 2.2, after the collision family (6.7), in Sections 9 and 10, and in Appendix C; this appendix is new. The PDF bookmark of the title page was fixed (no duplicate page anchor), and the command-line option `--output` in Appendix C is now typeset with two hyphens (the delivered PDF showed one dash).

No mathematical statement of the manuscript was changed and no symbol was renamed; the added notes write α for the project's coefficient $a = [t^2]p$, because a is a family parameter here. The report's $t = xy$ agrees with the project's research notes but not with the sibling report on arithmetic fibers, where $t = y + 1/x$.

D.2 Repository statements at the pin

[write] The source read the project README, its research notes and `verify_equivariant_sparsity.py` at its pin. The project did not change between the pin and the placement commit. Its statements about the project are accurate: the map, determinant and collision are the project's (and formalized, Section 1.4); the sparsity script proves that eleven coefficients cannot vanish and identifies the branch $q_{40} = 0$ (script lines 110–127 and 164–167); its certificate contains $(q_{21} - 4)^2$ with the comment that the square “records a harmless nonreduced structure” (lines 129–132), and neither README identifies the scheme as a double point. The one omission, the credit for the determinant identity and the collision mechanism, is repaired in Section 1.4. Shaska [4] and the Stacks Project references were not checked at the write; Gao [5] was checked for Theorems 3.3 and 3.5 (Section 1.5).

At the write, the ideal statement of Theorem 8.1 was also checked in both directions, which the shipped script does not do: with SymPy 1.14.0 the eighteen coefficient equations and the twelve generators of (A.1) have the same reduced Gröbner basis (grevlex), and each set lies in the ideal of the other. The normalization (3.4) was rechecked the same way.

D.3 Rerunning the checks

[write] The script writes its JSON record to `--output`, default `verification.json` in the current directory. Run it from the report with an output path outside the repository, so that neither the report root nor `data/` is written:

```
python code/verify_results.py --output <scratch>/verification.json
```

It needs Python 3.10 or later and SymPy (data/requirements.txt pins `sympy==1.14.0`). At the write it was run so with Python 3.13.5 and SymPy 1.14.0: all thirteen check groups passed, the JSON record equals `data/verification.json` apart from line endings, and the printed output equals `data/verification.txt` apart from line endings and the certificate path in its last line. The record embeds the Python version, so a run under another version differs in that field.

References

- [1] Vladimir Reshetnikov, *ProveIt: Algebra/JacobianConjecture/README.md*. Repository snapshot e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9, inspected September 24, 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/README.md>.
- [2] Vladimir Reshetnikov, *ProveIt: Algebra/JacobianConjecture/Research/README.md*. Same repository snapshot and inspection date. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/Research/README.md>.
- [3] Vladimir Reshetnikov, *verify_equivariant_sparsity.py*. Same repository snapshot and inspection date. https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/Research/verify_equivariant_sparsity.py.

- [4] T. Shaska, *Graded Keller maps and the Jacobian Conjecture*, arXiv:2607.20210, 2026. Version 2 dated July 25, 2026; consulted September 24, 2026. <https://arxiv.org/abs/2607.20210>.
- [5] Shuhong Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222, 2026. Version 1 dated July 31, 2026; consulted September 24, 2026. <https://arxiv.org/abs/2608.00222>.
- [6] The Stacks Project Authors, *The Stacks Project*, Theorem 10.34.1, Hilbert Nullstellensatz, Tag 00FV. Consulted September 24, 2026. <https://stacks.math.columbia.edu/tag/00FV>.
- [7] The Stacks Project Authors, *The Stacks Project*, Lemma 10.20.1, Nakayama's lemma, Tag 00DV. Consulted September 24, 2026. <https://stacks.math.columbia.edu/tag/00DV>.